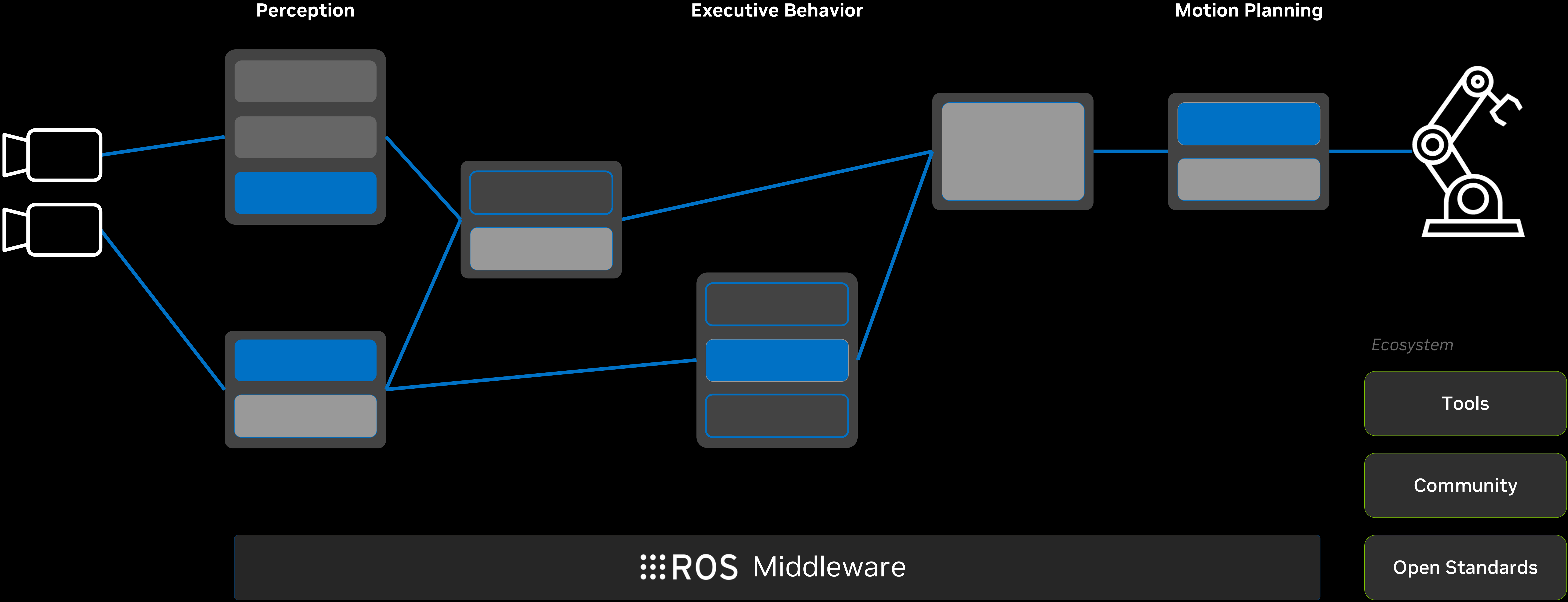




Accelerating ROS 2 Workloads With NVIDIA GPU-Powered Libraries and AI Models

What Is ROS?

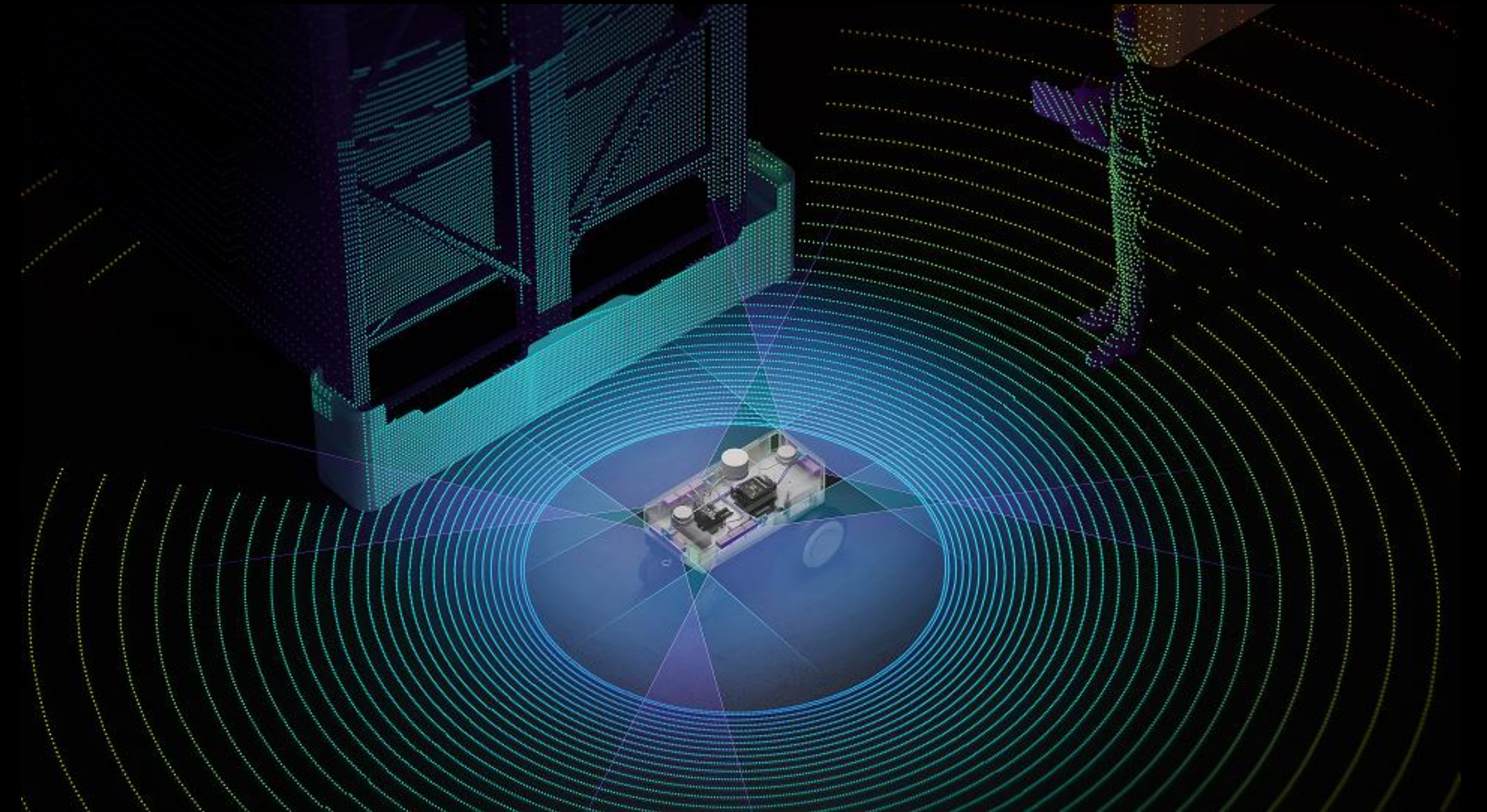
Robot Operating System



Powering AI-Enabled Robots With Hardware Acceleration

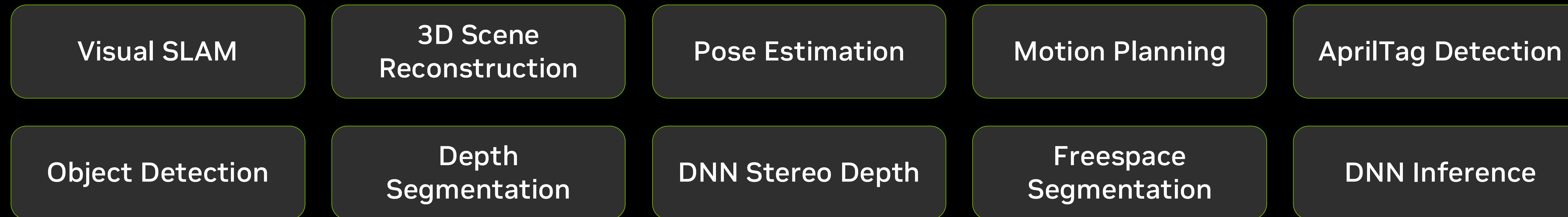
As robots advance and become more autonomous, they need **situational awareness** and **real-time responsiveness**.

Demanding workloads require more sensors and advance algorithms.



What Is NVIDIA Isaac ROS?

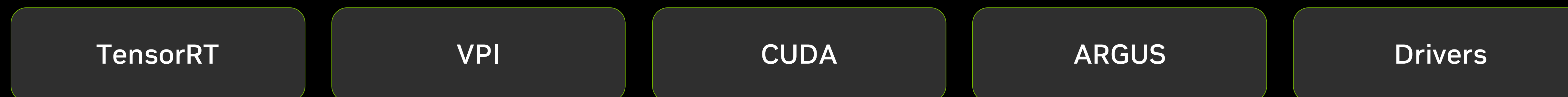
30+ Accelerated ROS 2 Packages



Integration



Accelerated Libraries



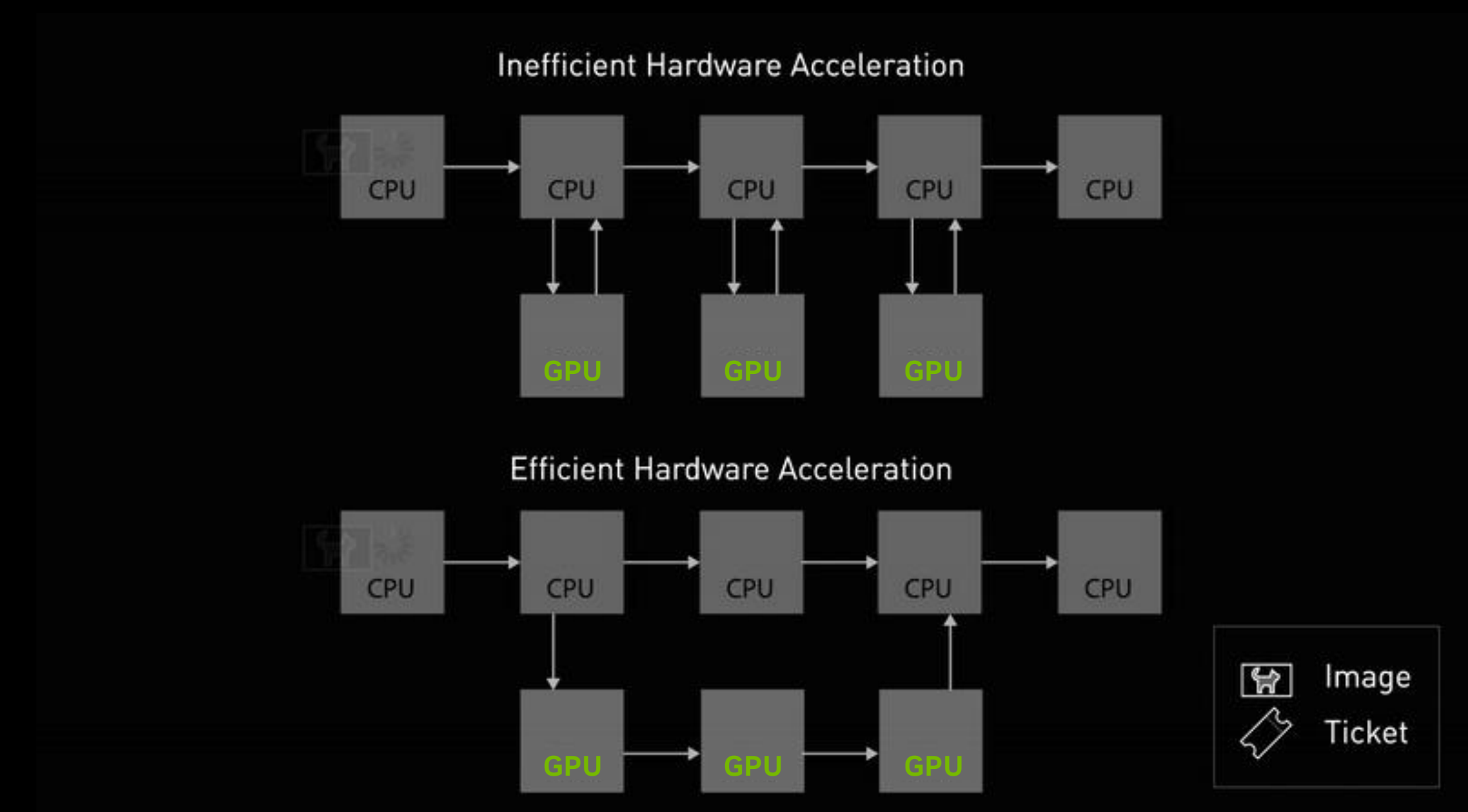
NVIDIA Isaac ROS is a collection of **NVIDIA CUDA-accelerated libraries** and **AI models**.

What Is NITROS?

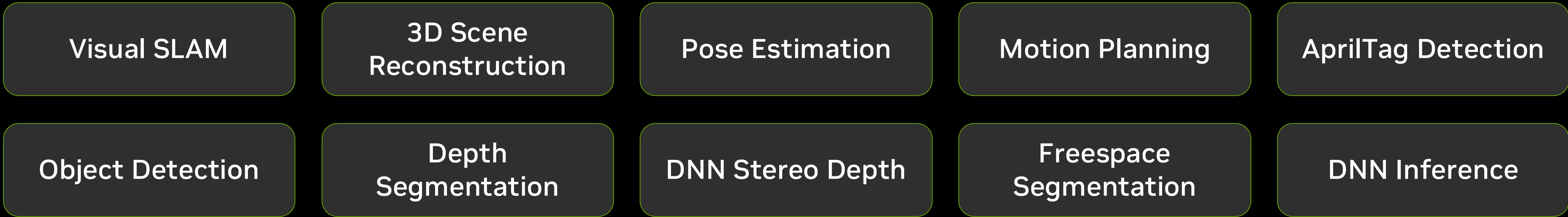
NVIDIA Isaac Transport for ROS

- NITROS avoids cycling memory back from accelerator caches to CPU memory with handles.
- Optimal when possible, compatible when necessary.
- Many technologies to enable developers to build their own NITROS-enabled nodes:
 - CUDA with NITROS
 - NITROS Bridge
 - PyNITROS

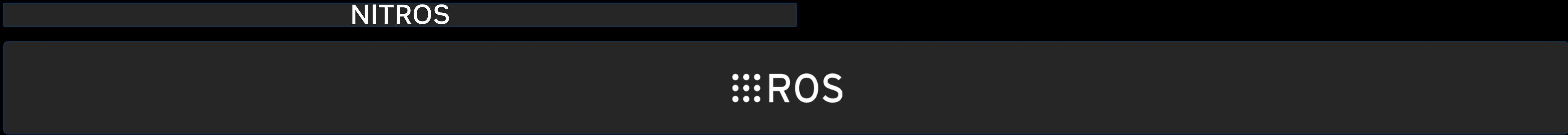
NITROS powered ROS 2 Graphs



30+ Accelerated ROS 2 Packages



Integration

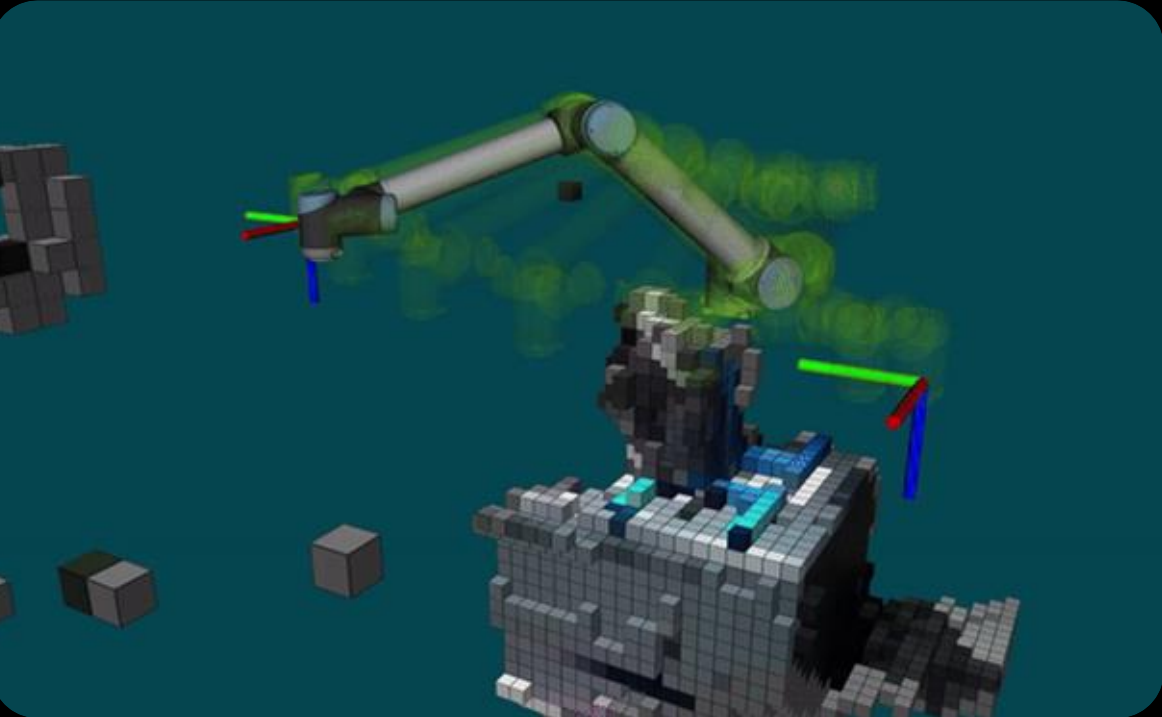


Accelerated Libraries

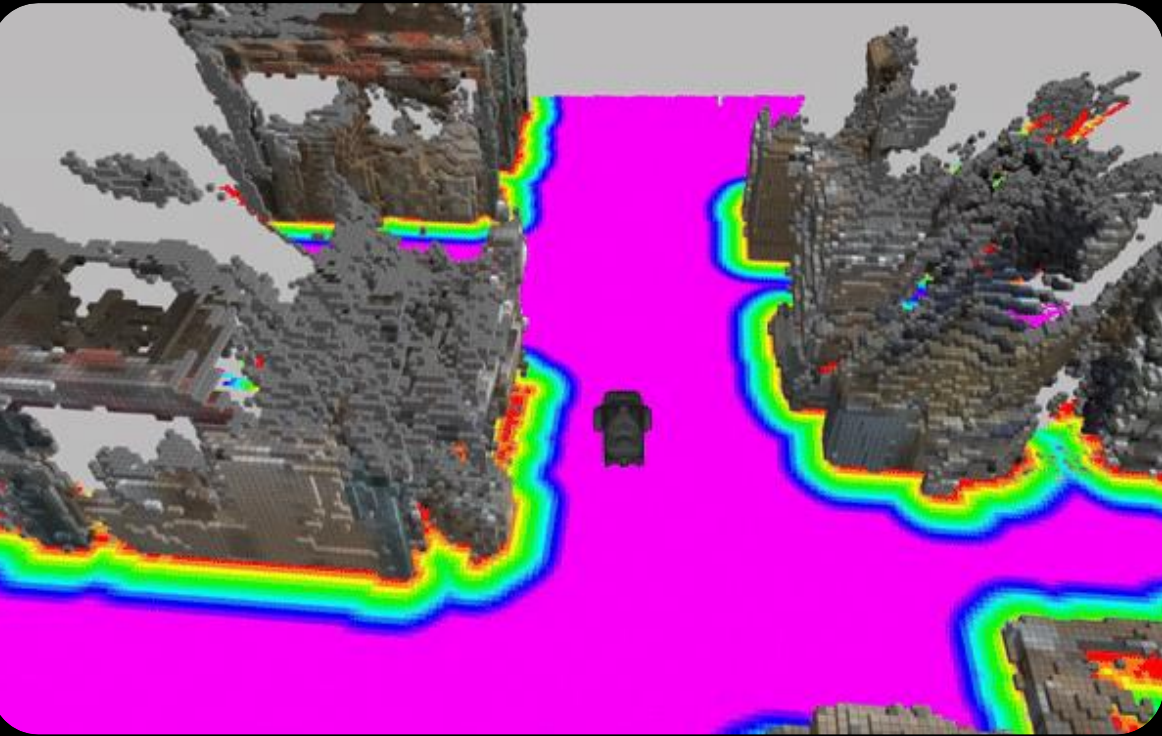


Reference Workflows

Isaac Manipulator



Isaac Perceptor



NVIDIA Isaac Perceptor

Reference workflows for developing autonomous mobile robots



Up to 4 Stereo Cameras

Isaac Perceptor



Disparity

Visual Odometry

People Segmentation

3D Occupancy Grid

ROS

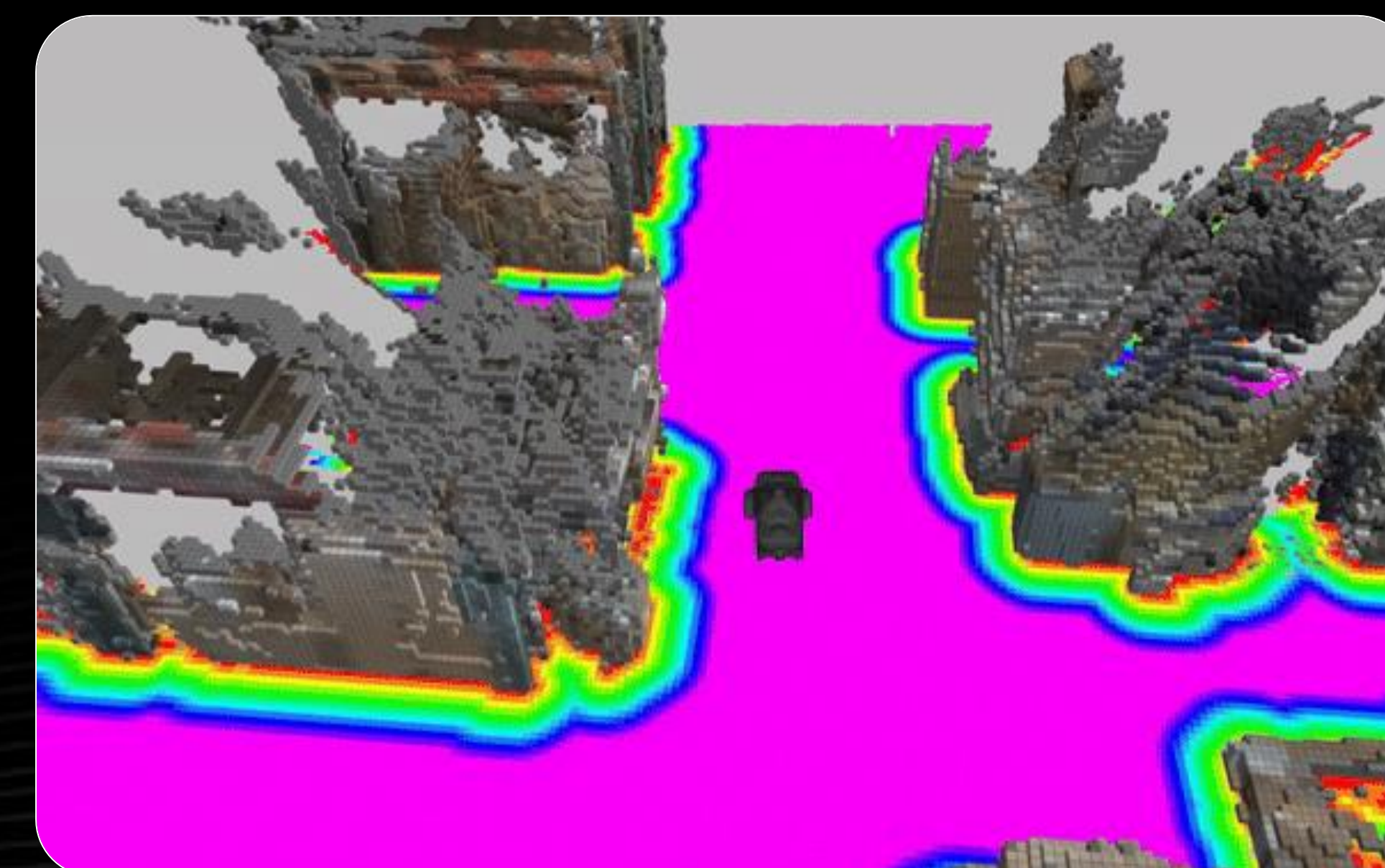
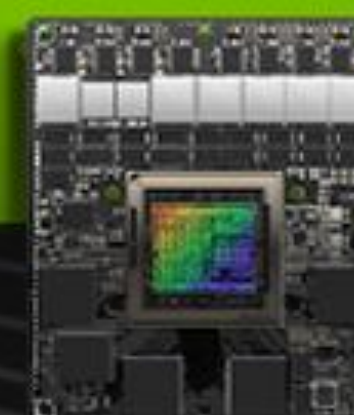
Isaac ROS

AI Acceleration

30+ ROS Packages

Fully Optimized

NVIDIA Nova Orin



NVIDIA Isaac Manipulator

Reference workflows for developing autonomous manipulation



Isaac Manipulator



6D Pose Tracking

Motion Planning

Grasp Identification

Manipulation &
Dexterity

ROS

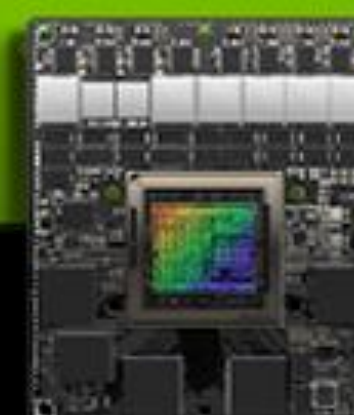
Isaac ROS

AI Acceleration

30+ ROS Packages

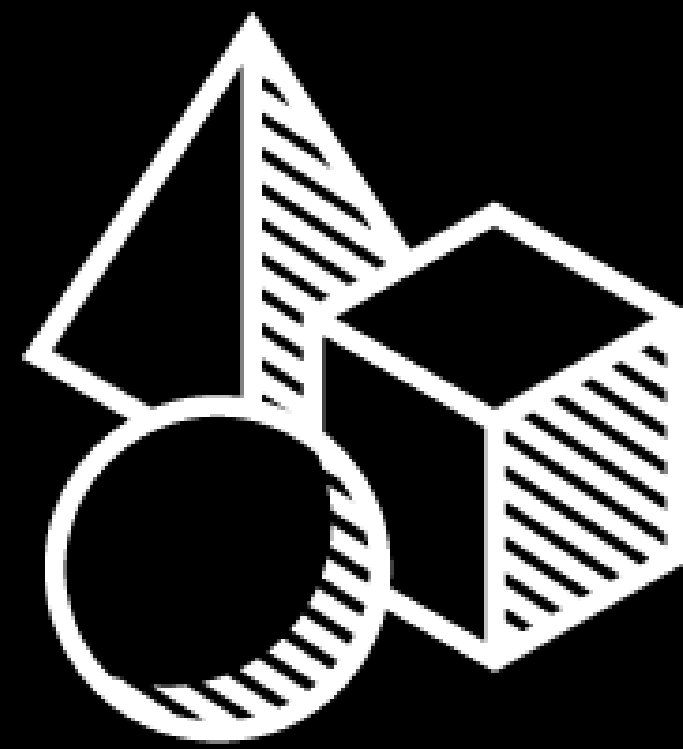
Fully Optimized

NVIDIA Jetson
AGX Orin

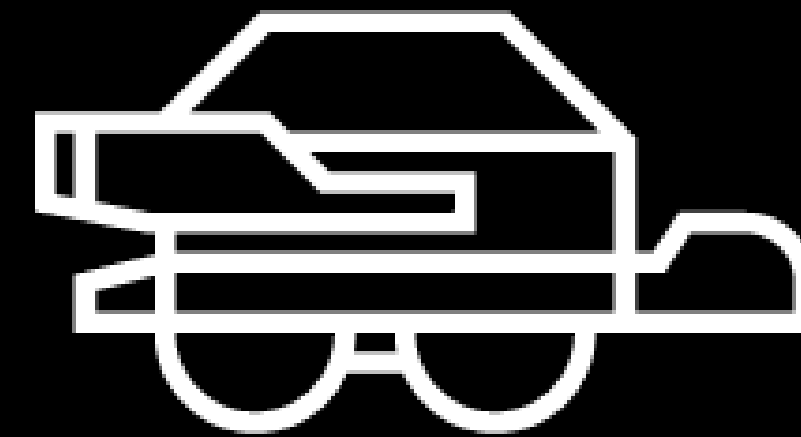


Lab Objectives

Hands-on: Perception Driven Pick-And-Place

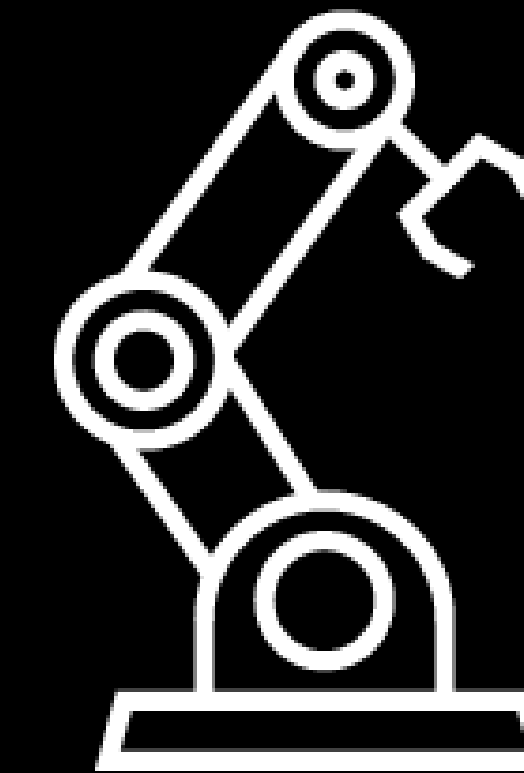


**Understand the
fundamental Isaac ROS
building blocks**



**Learn about hardware
acceleration libraries
and AI models**

cuMotion
FoundationPose
SyntheticaDeTR
nvBlox



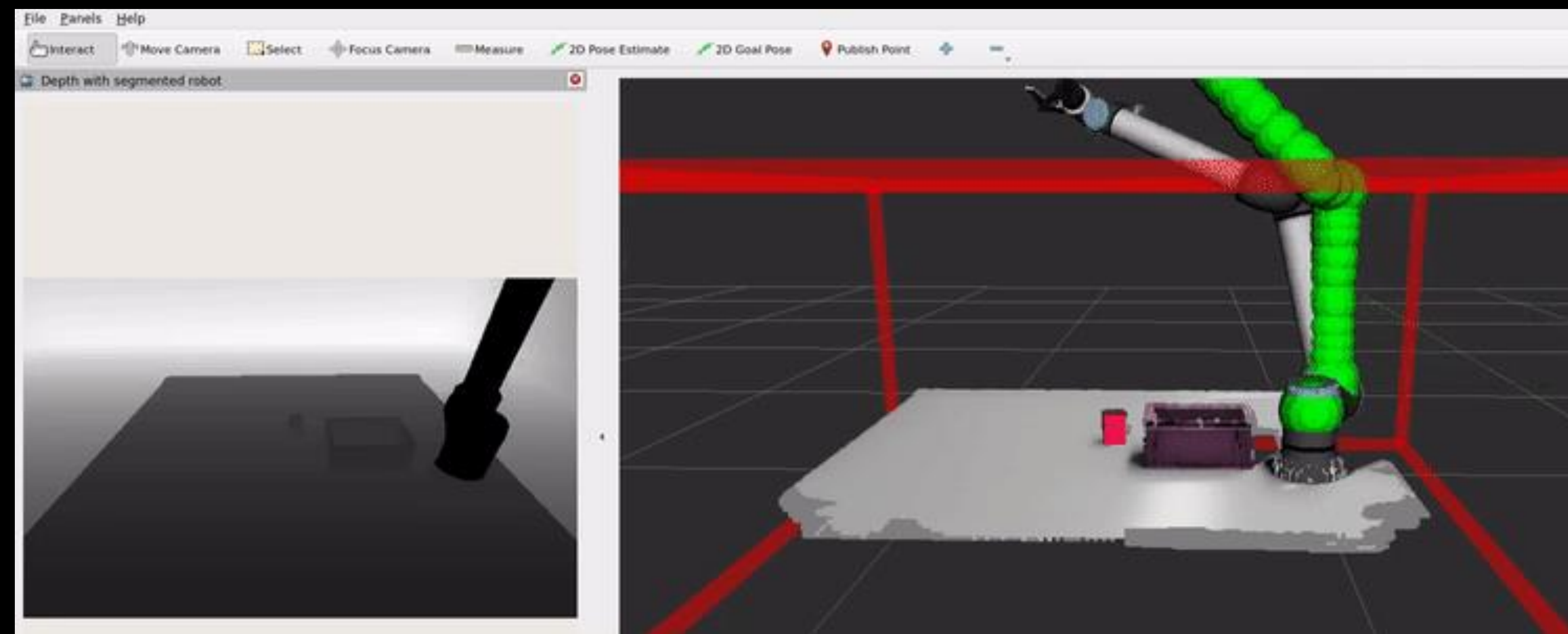
**Build an end-to-end
perception-driven pick
and place workflow**



Isaac Manipulator

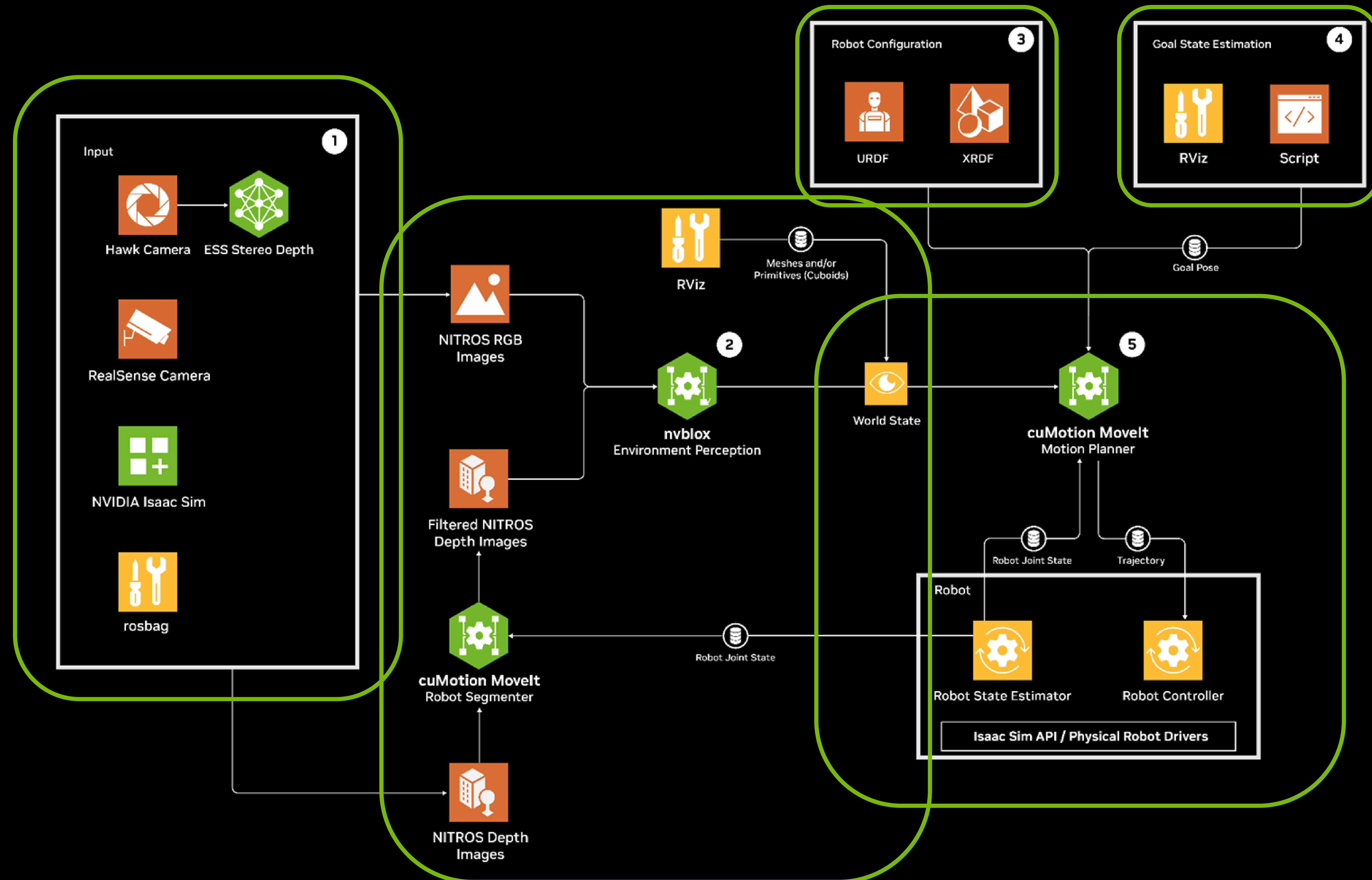
Isaac Manipulator In Action

What we're doing today: pick and place with obstacle avoidance



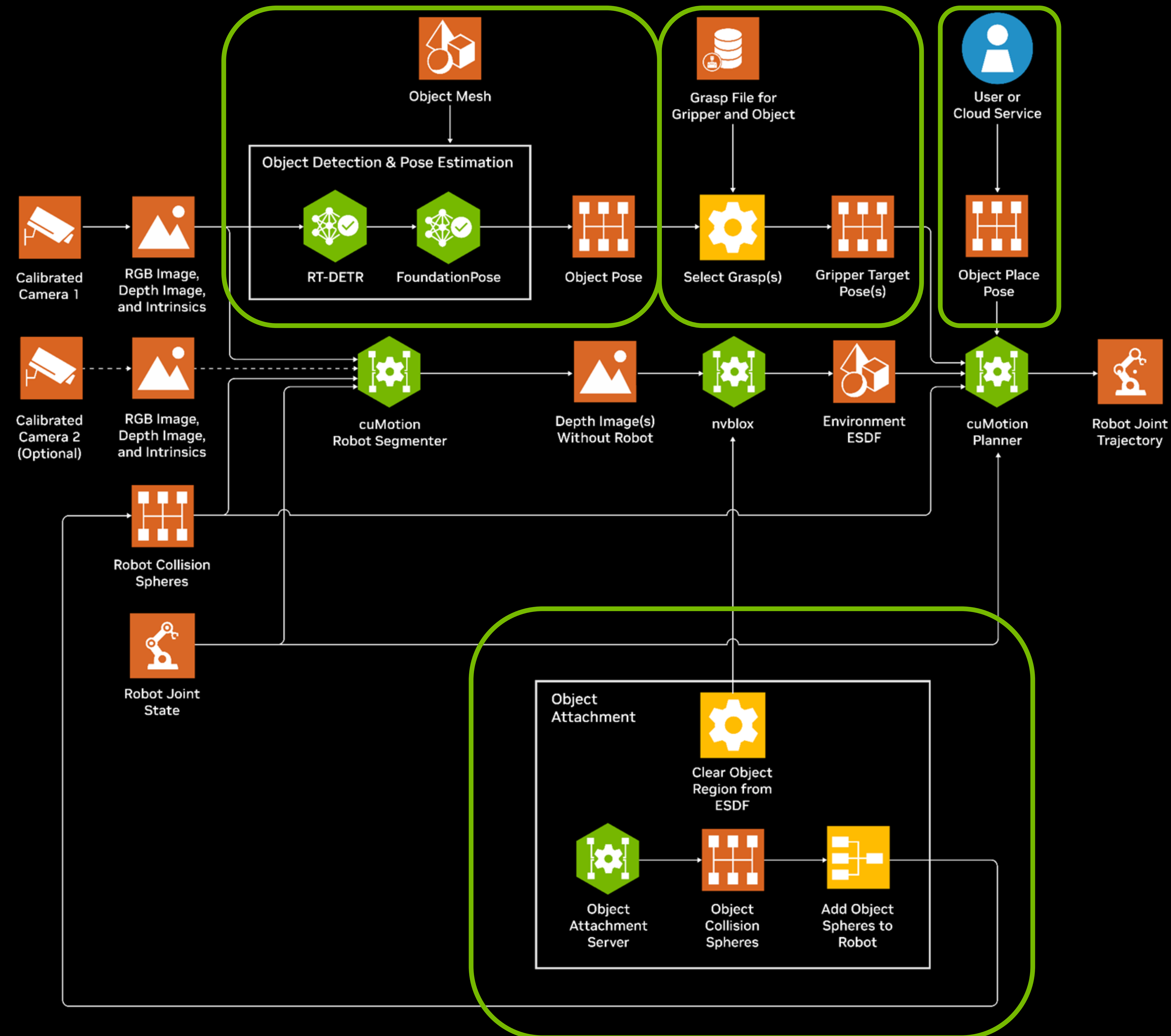
Isaac Manipulator

Reference Architecture



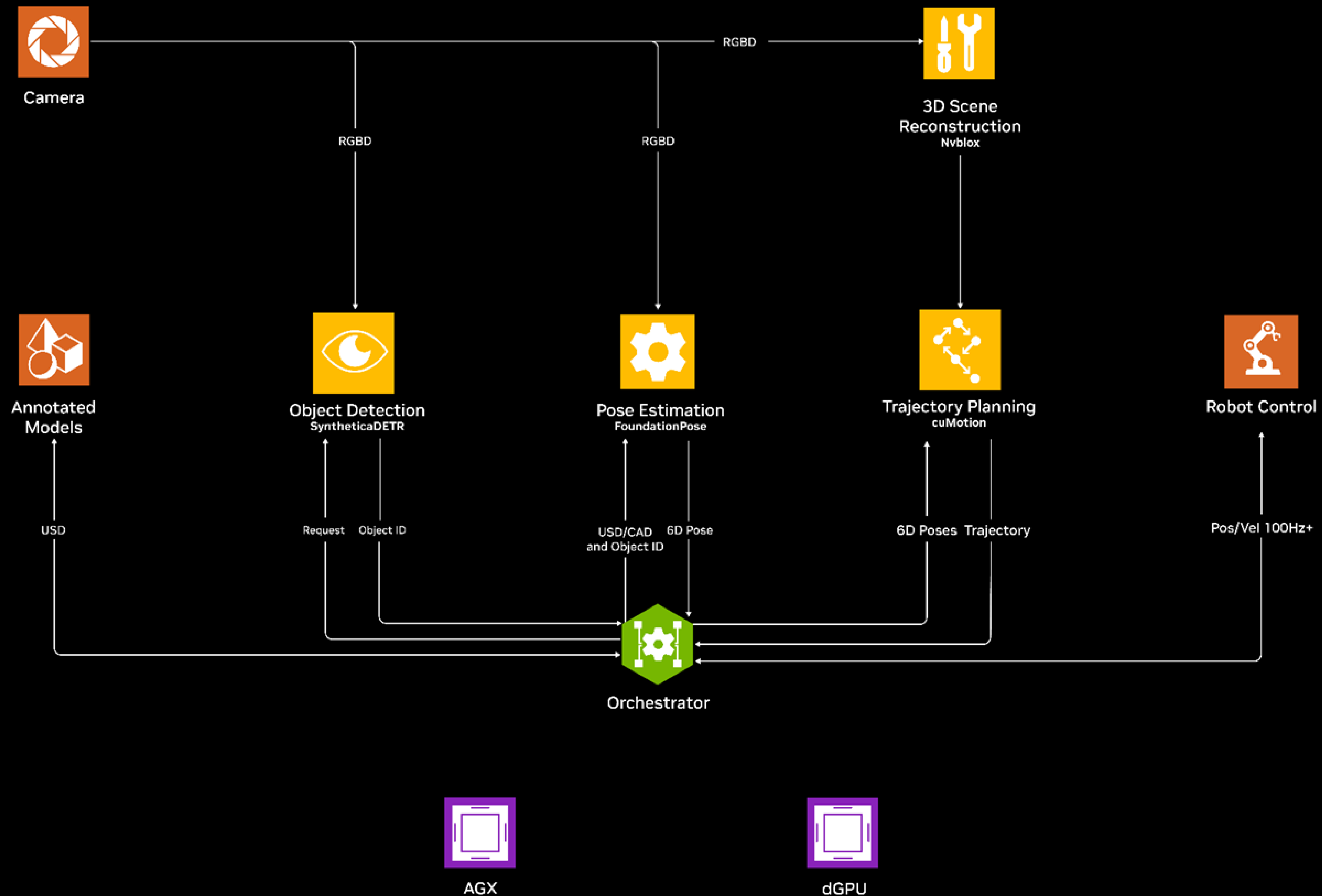
Pick And Place

Architecture for a specific Isaac Manipulator workflow



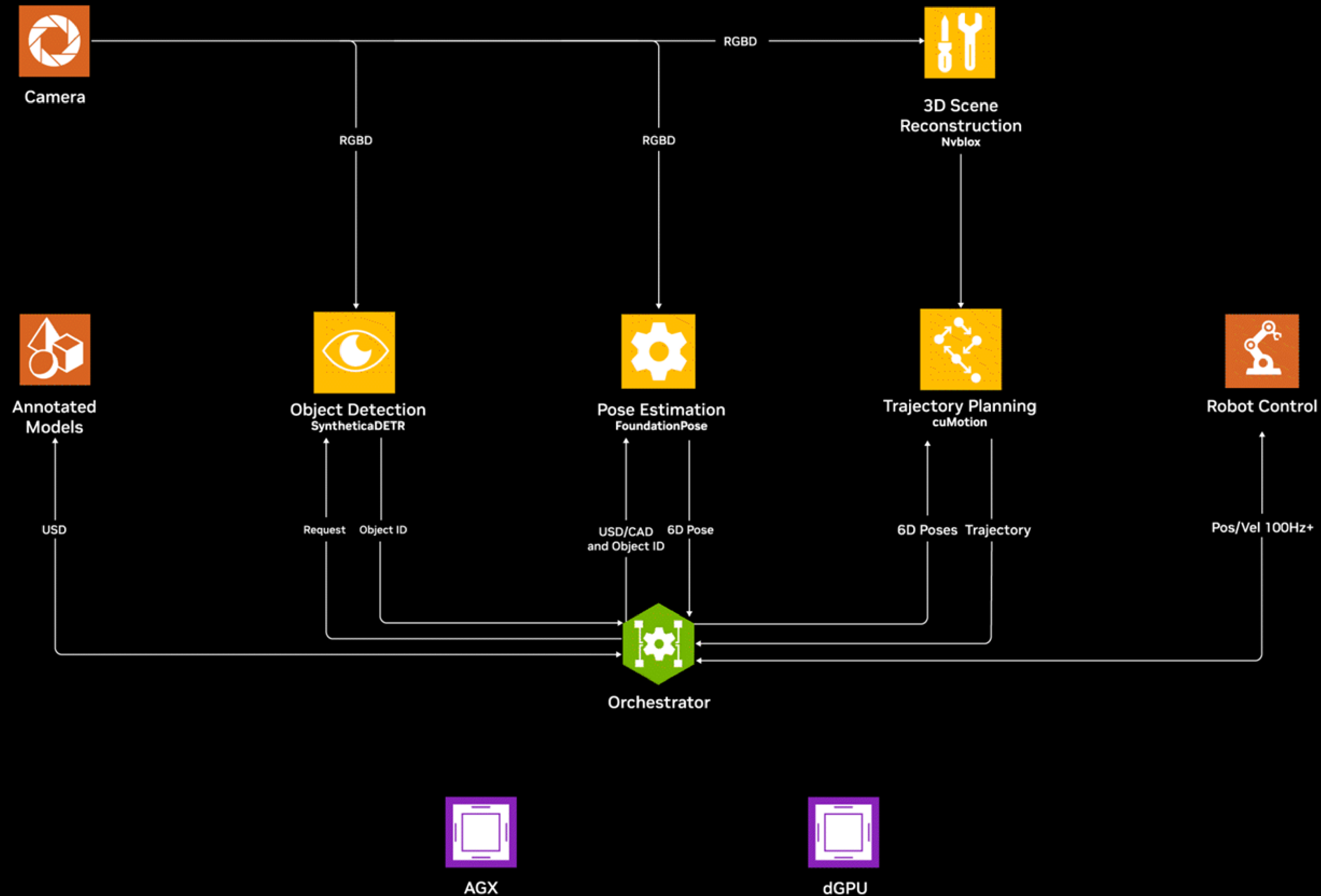
Pick And Place Orchestration

Orchestrator: the high level control of Pick and Place workflow



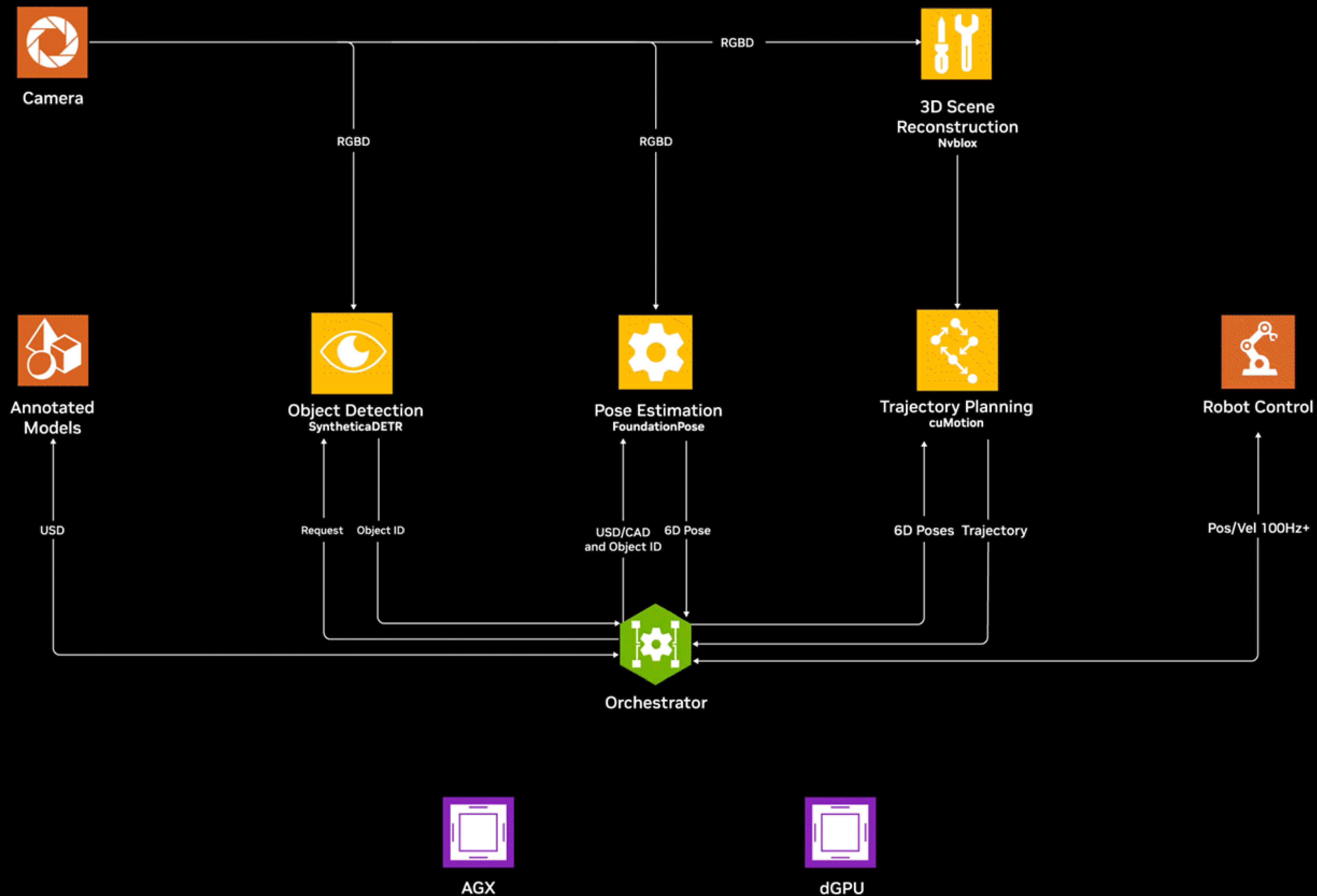
Pick And Place Orchestration

Orchestrator: the high level control of Pick and Place workflow



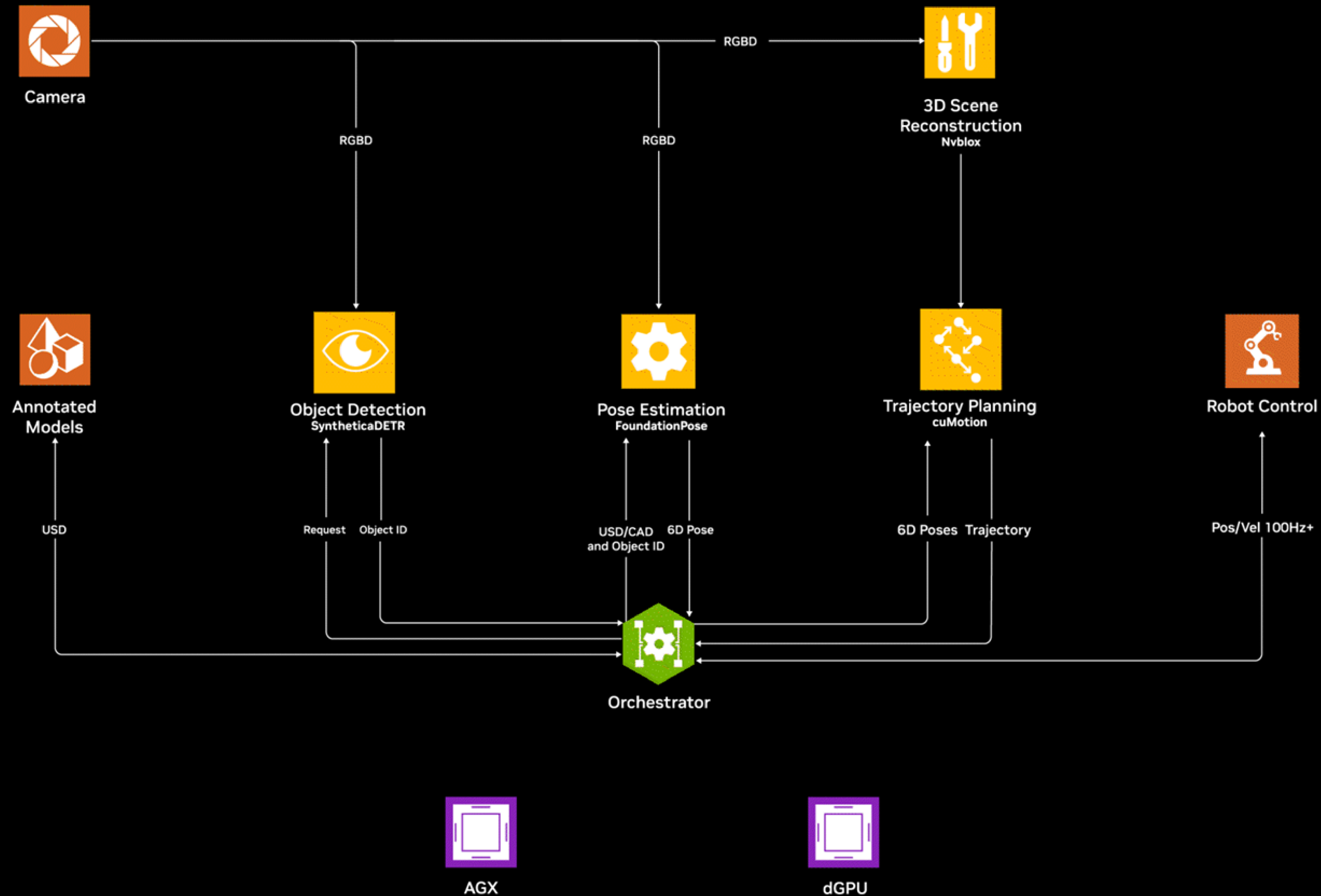
Pick And Place Orchestration

Orchestrator: the high level control of Pick and Place workflow



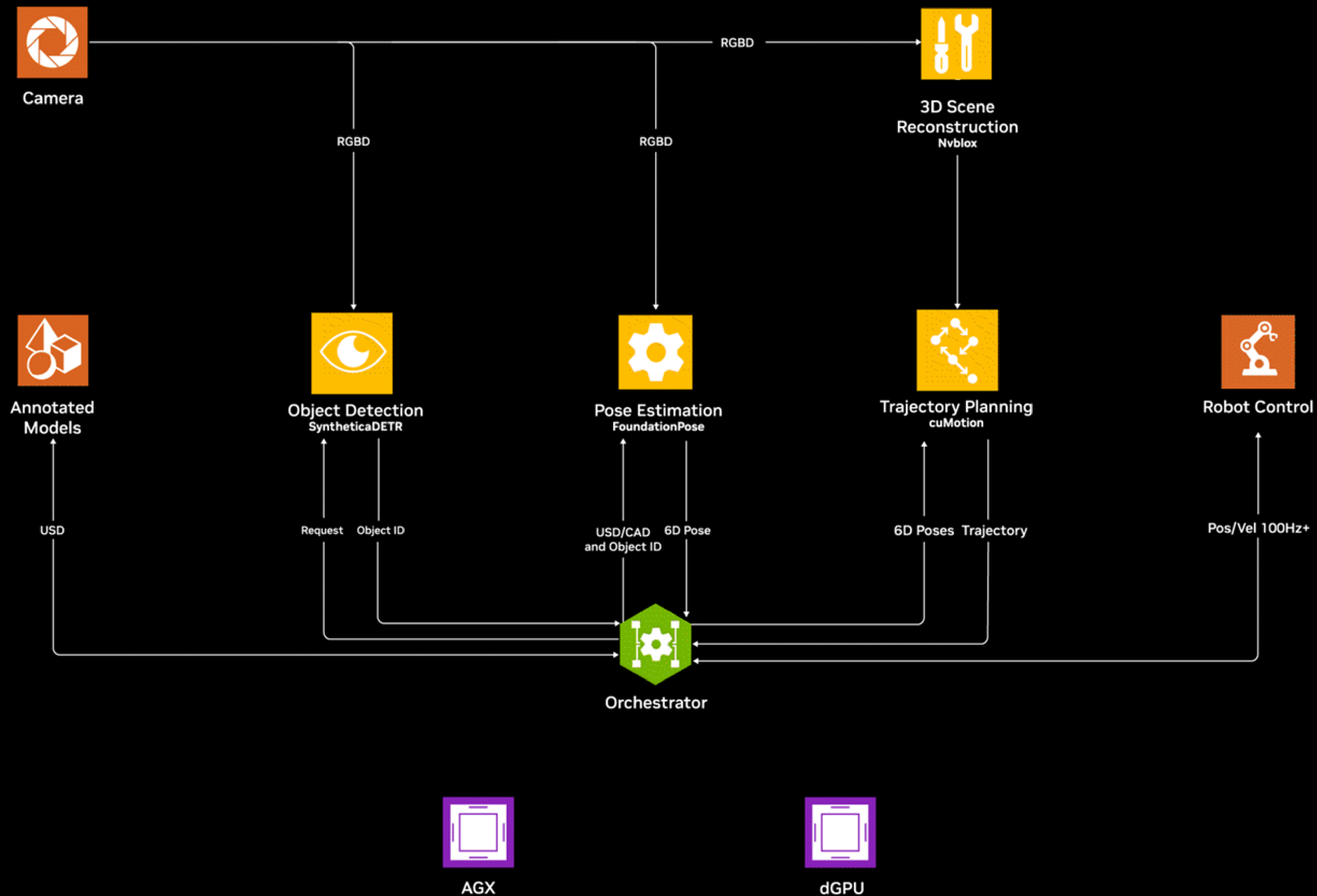
Pick And Place Orchestration

Orchestrator: the high level control of Pick and Place workflow



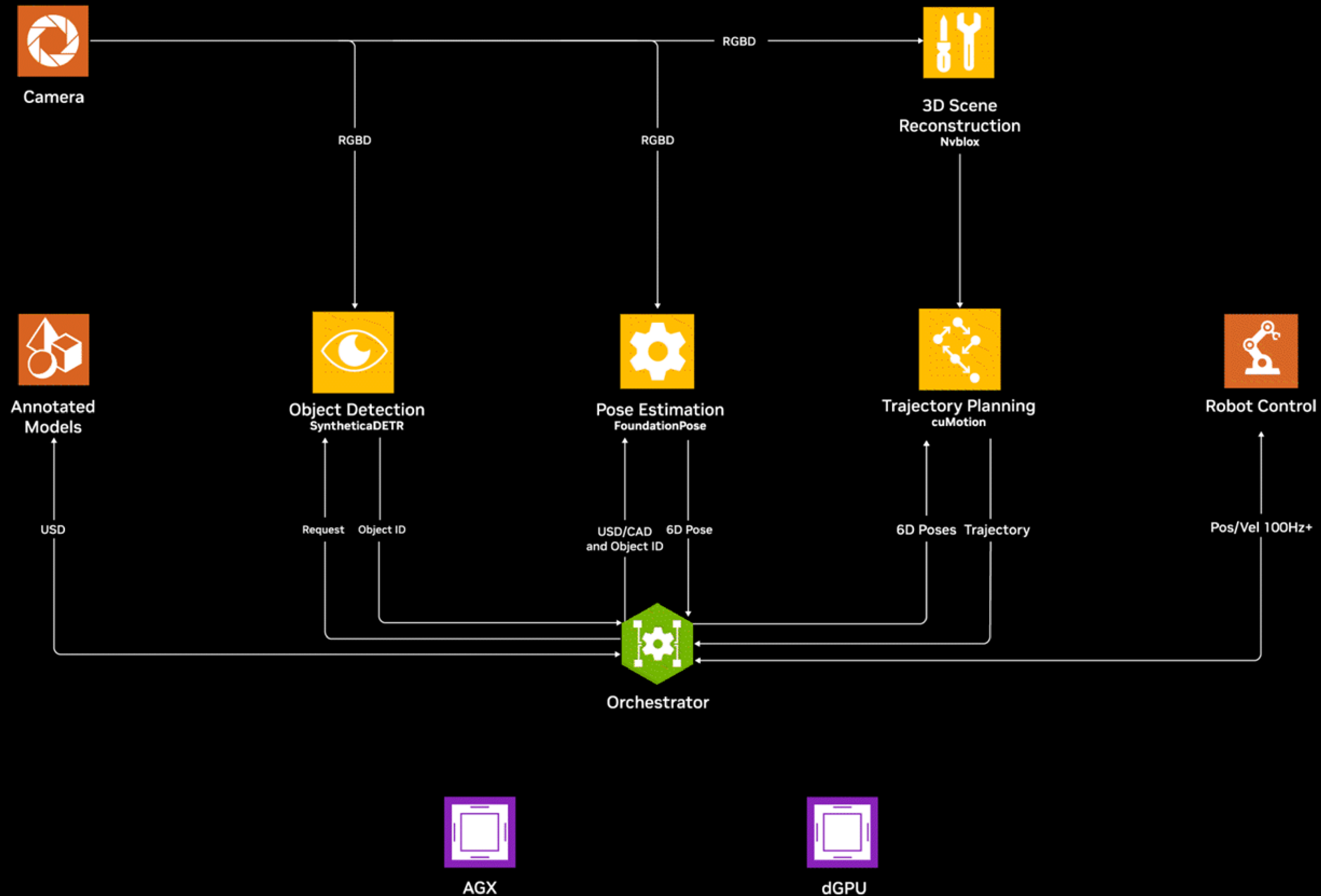
Pick And Place Orchestration

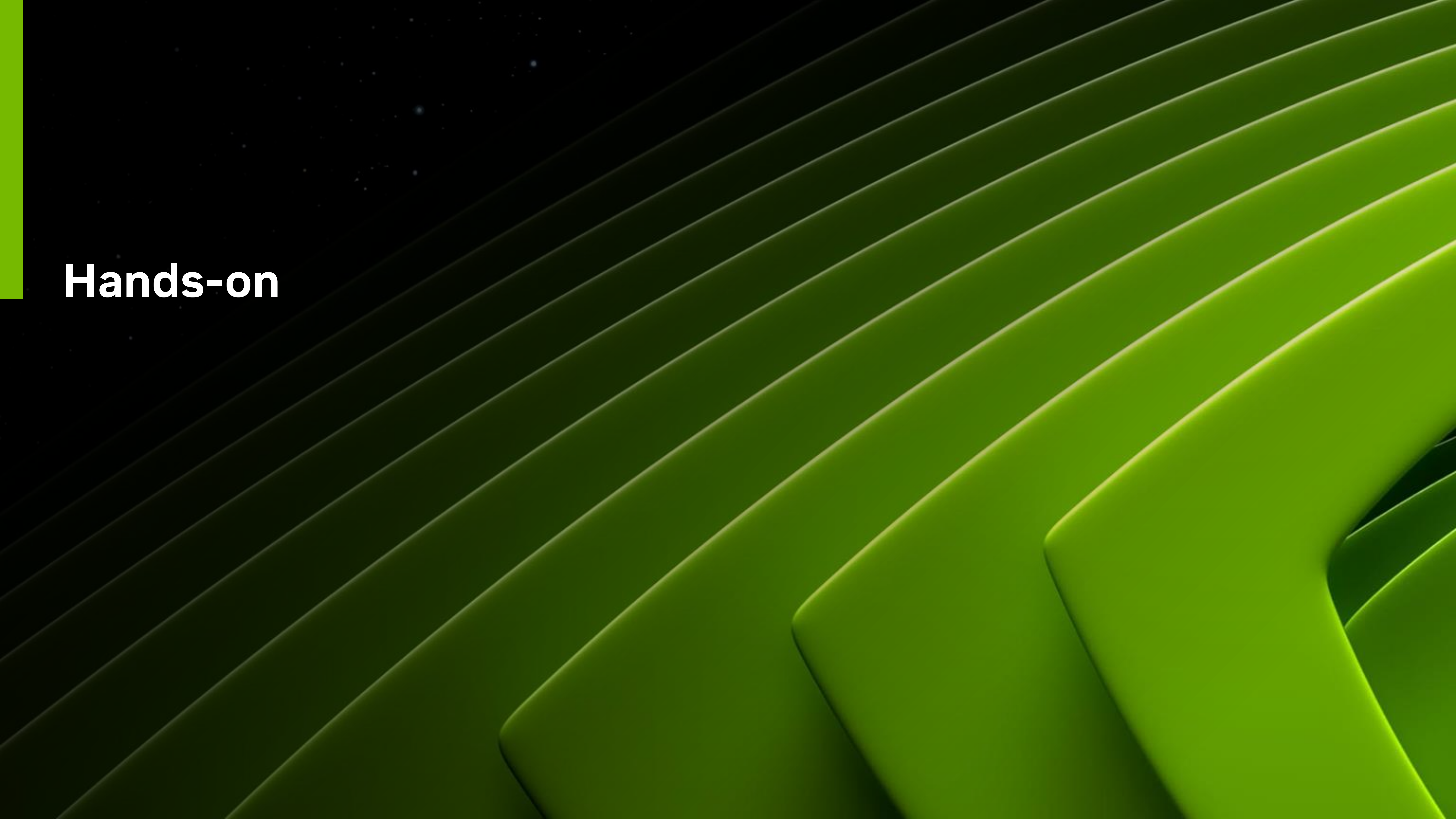
Orchestrator: the high level control of Pick and Place workflow



Pick And Place Orchestration

Orchestrator: the high level control of Pick and Place workflow





Hands-on

Development Environment

Overview of our setup for today

The following has been preconfigured for you:

- ✓ Two systems:
 - One for Isaac Sim - interact with it using GUI
 - One for Compute stack with Isaac ROS - interact with it using terminals
- ✓ Pre-trained models
 - RT_DETR - Object detection
 - Foundation Pose - Pose Estimation
 - Already converted in the form ready to be run with TensorRT (optimization of the onnx model)

Common Commands

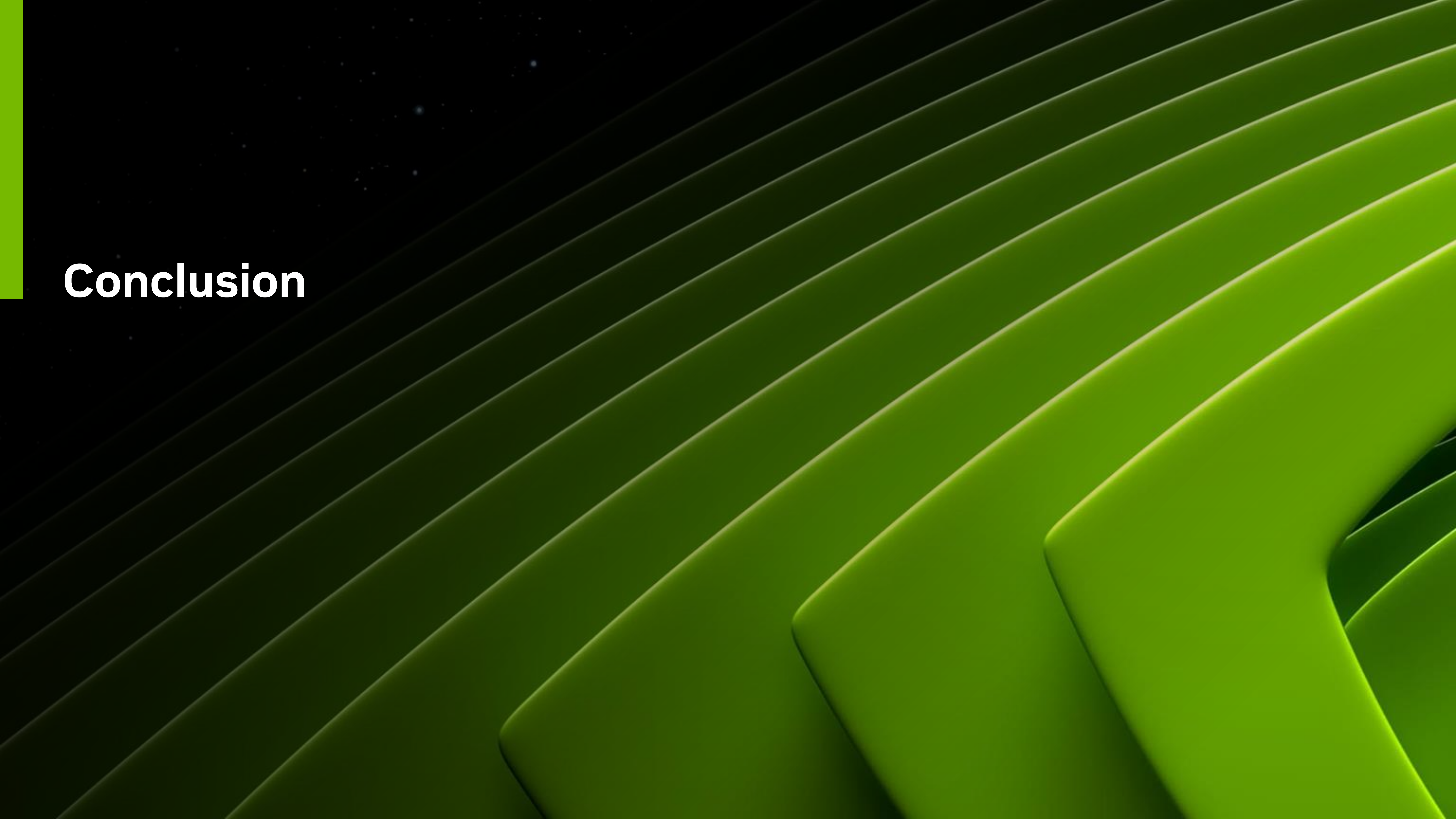
Everytime we create new terminal in Isaac ROS system

1. Isolate ROS network with your computer ID number to avoid cross-talk

```
export ROS_DOMAIN_ID=<YOUR_ID>
```

1. Source the ROS workspace

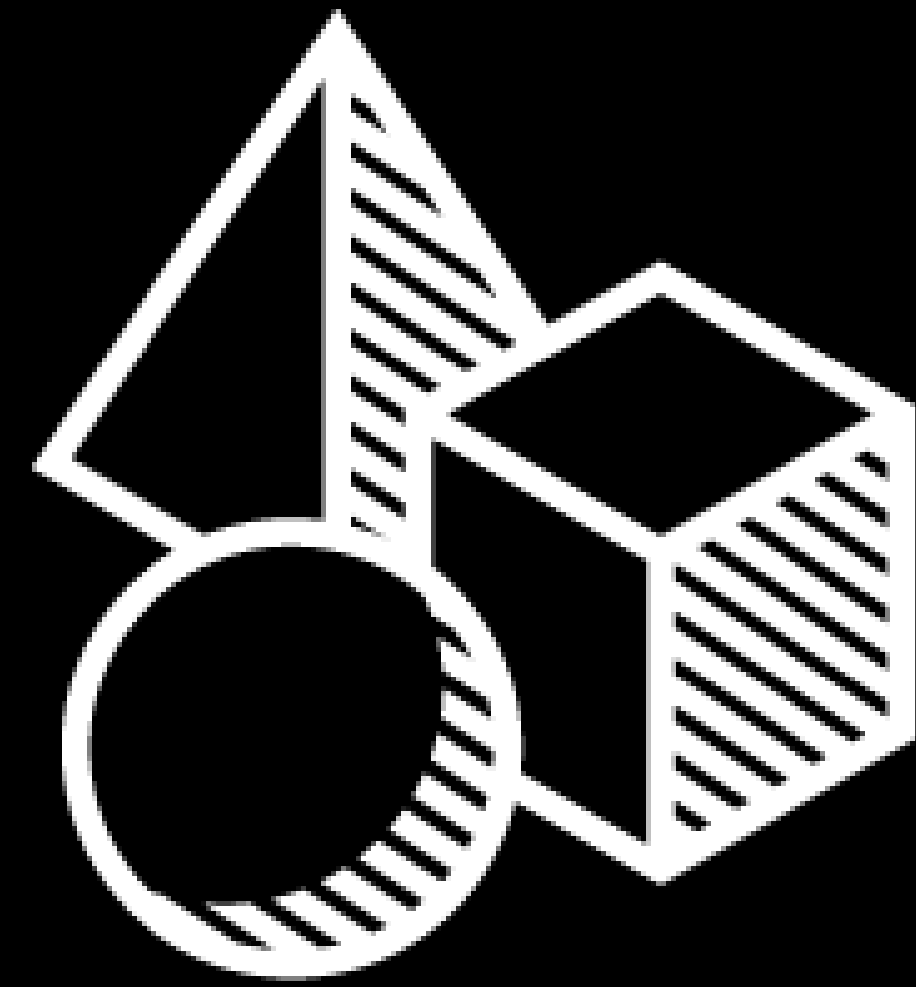
```
cd $ISAAC_ROS_WS && source install/setup.bash
```

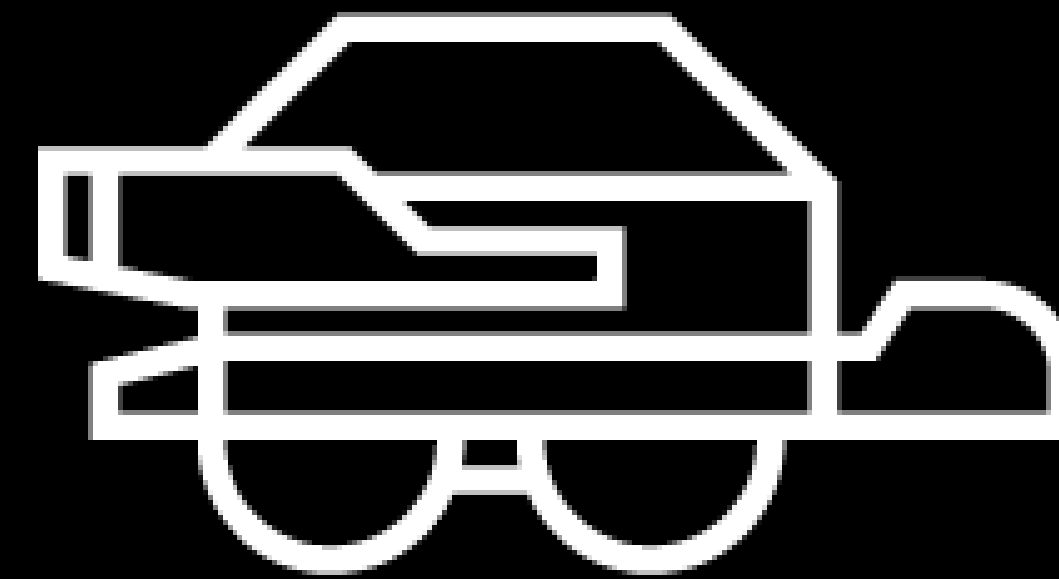
Conclusion

Review

What did we learn today?

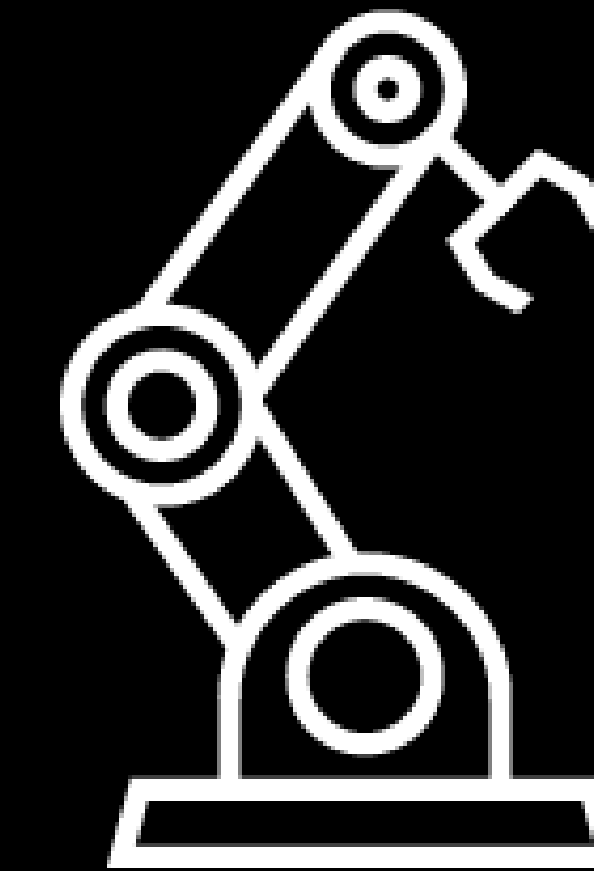


**Gained understanding of
basic NVIDIA Isaac ROS
building blocks**



**Learned about hardware
acceleration libraries
and AI models**

cuMotion
FoundationPose
SyntheticaDeTR
nvBlox



**Built an end-to-end
perception driven pick
and place workflow**

Examples of Isaac ROS in the real world



Teradyne Robotics



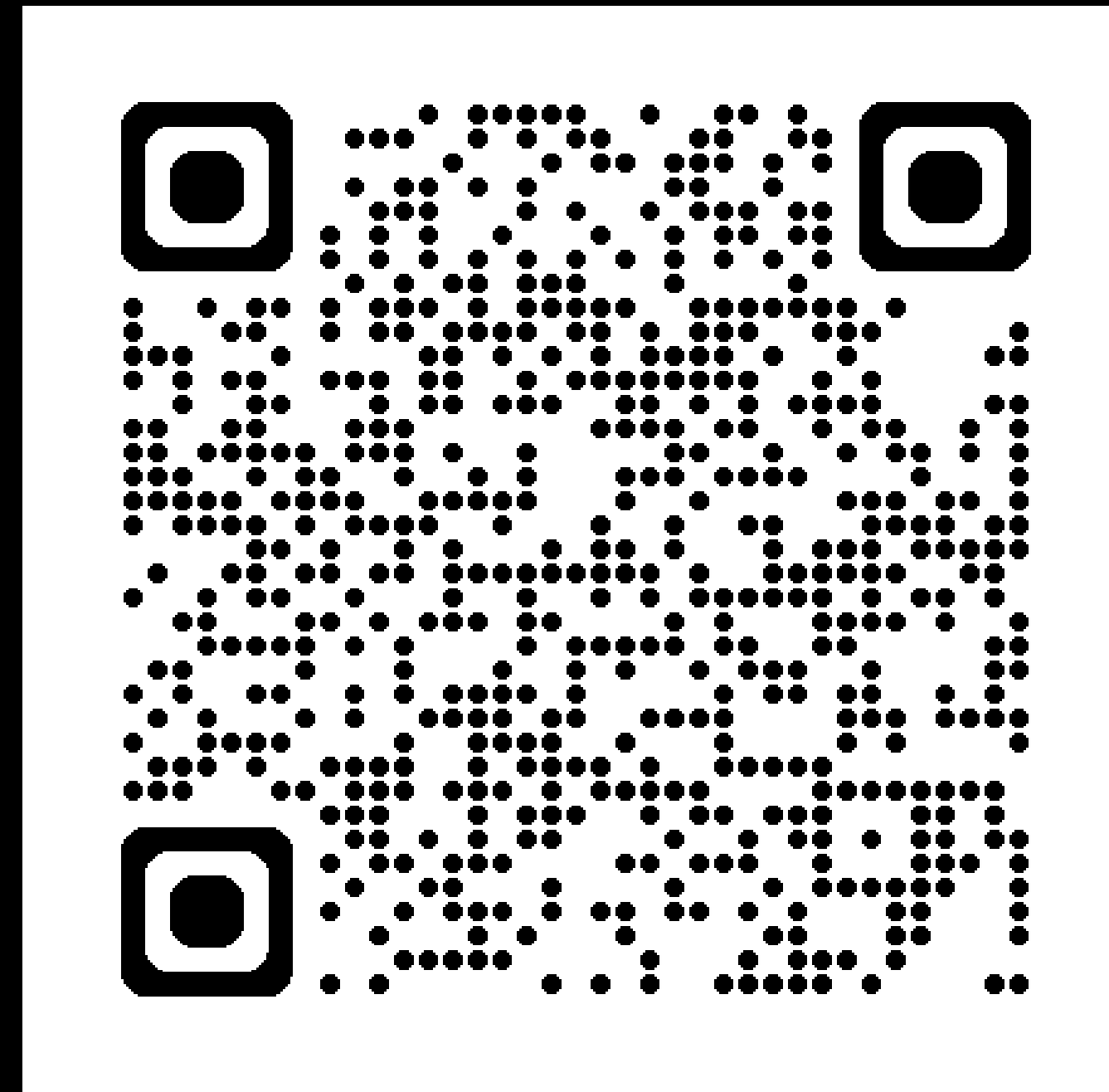
Vention



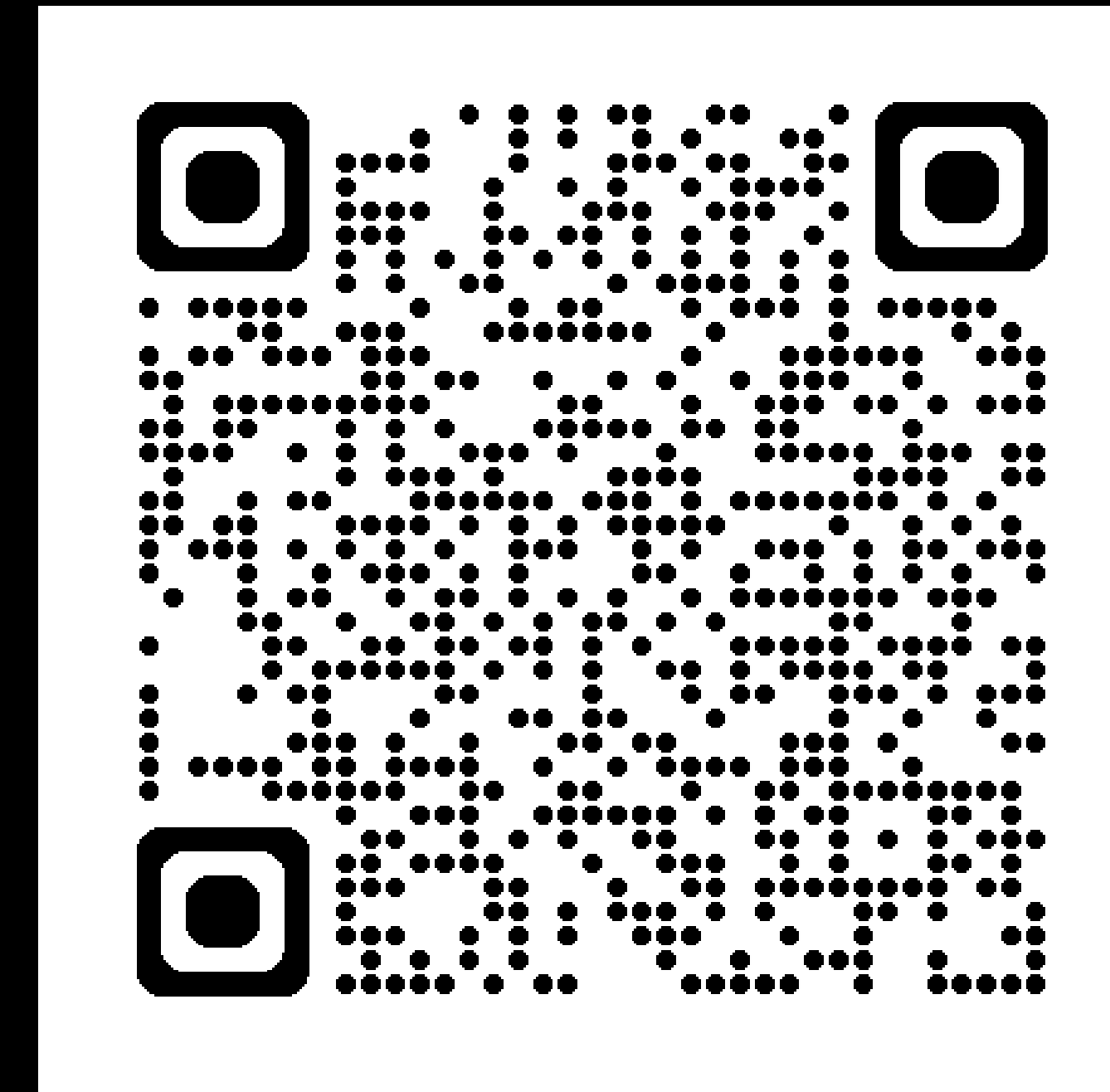
Miso Robotics

Get Started With Isaac ROS Today!

Hardware acceleration for today's robots



Tutorials, concepts and more >
[Documentation](#)



Stay in touch > [Forums](#)

